

Bus Reservation System is a Java console-based application designed to manage bus ticket bookings efficiently.

It allows an Admin to manage buses and a Passenger to book seats while ensuring proper validation and availability and managing the system securely.

The system allows users to view available buses, book tickets, cancel bookings, and check booking details.

- 1) User (Abstract Class)
 - Base class for system users
 - Stores common details like `name` and `phone`
 - Declares abstract method `displayInfo()`
- 2) Admin (extends User)
 - Manages buses (add/view buses)
 - Has higher privileges than passengers
- 3) Passenger (extends User)
 - Used to book tickets
 - Displays passenger details
- 4) Bus
 - Stores bus details such as bus number and available seats
 - Manages seat count securely
- 5) Booking
 - Represents a ticket booking
 - Connects Passenger and Bus objects
- 6) AvailabilityChecker
 - Checks whether a bus exists
 - Verifies seat availability
- 7) MainSystem
 - Entry point of the application
 - Handles user interaction and menu-driven flow

- 1) Encapsulation
 - Bus class protects `seatCount`
 - Accessed only through methods like `getSeatCount()` and `bookSeats()`

Prevents unauthorized modification of seat data.

2) Inheritance

- Admin and Passenger inherit from User
Avoids code duplication and improves reusability.

3) Polymorphism

- displayInfo() method
Allows flexibility and dynamic method execution.

4) Abstraction

- User is an abstract class
Forces subclasses to implement required behavior.

PS/2023/232 (M.R.Rishad Ahamed)

- Explain system purpose
- Describe overall workflow
- Doing a basic classes

PS/2023/277(T.Vakeeshan)

- Add extra 3 classes for more realistic
- After do a small modification

PS/2023/021(R.Ravikumar)

- Adding two OOP principles
- After do a small modification

PS/2023/400(M.A.Ahamed)

- Adding two OOP principles
- After do a small modification

PS/2023/325(T.Gajanan)

- Modify All the classes for more realistic
- Help everyone for our presentation
- Doing this final report

